In [1]:
```
import numpy as np
import pandas as pd

from sklearn.datasets import fetch_20newsgroups
from sklearn.feature_extraction.text import CountVectorizer

# 1. Load the 20 Newsgroups training dataset
print("Loading 20 Newsgroups dataset...")

data = fetch_20newsgroups(
    subset="train",
    remove=("headers", "footers", "quotes"),
    random_state=42
)

print("Total documents:", len(data.data))
print("Number of categories:", len(data.target_names))

# 2. Convert text documents into word-count vectors
vectorizer = CountVectorizer(
    stop_words="english",
    max_features=3000,
    min_df=2,
    max_df=0.8
)

X = vectorizer.fit_transform(data.data)
feature_names = vectorizer.get_feature_names_out()

print("Document-term matrix shape:", X.shape)
print("Vocabulary size:", len(feature_names))

# 3. Select documents belonging to sci.space
category_name = "sci.space"
category_id = data.target_names.index(category_name)

category_indices = np.where(data.target == category_id)[0]
category_word_counts = np.asarray(
    X[category_indices].sum(axis=0)
).ravel()

print("\nSelected category:", category_name)
print("Number of documents:", len(category_indices))
print("Total word occurrences:",
      int(category_word_counts.sum()))

# 4. MLE estimation
total_words = category_word_counts.sum()

if total_words == 0:
    raise ValueError("No word occurrences found in the selected category.

mle_probabilities = category_word_counts / total_words

print("\nMLE estimation completed.")

# 5. MAP estimation with different symmetric Dirichlet priors
# Alpha must be >= 1 for the mode formula used here.
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```
prior_values = [1.0, 1.1, 2.0, 10.0]

map_estimates = {}

vocabulary_size = len(feature_names)

for alpha in prior_values:
    numerator = category_word_counts + alpha - 1
    denominator = total_words + vocabulary_size * (alpha - 1)

    map_probabilities = numerator / denominator
    map_estimates[alpha] = map_probabilities

print("MAP estimation completed for all prior values.")

# 6. Compare the estimated distributions
comparison_results = []

def calculate_entropy(probabilities):
    positive_probabilities = probabilities[probabilities > 0]
    return -np.sum(
        positive_probabilities * np.log2(positive_probabilities)
    )

comparison_results.append({
    "Method": "MLE",
    "Prior alpha": "None",
    "Zero probabilities": int(np.sum(mle_probabilities == 0)),
    "Entropy (bits)": calculate_entropy(mle_probabilities),
    "L1 distance from MLE": 0.0
})

for alpha, probabilities in map_estimates.items():
    comparison_results.append({
        "Method": "MAP",
        "Prior alpha": alpha,
        "Zero probabilities": int(np.sum(probabilities == 0)),
        "Entropy (bits)": calculate_entropy(probabilities),
        "L1 distance from MLE": np.sum(
            np.abs(probabilities - mle_probabilities)
        )
    })

results_df = pd.DataFrame(comparison_results)

print("\nCOMPARISON OF MLE AND MAP ESTIMATES")
print(results_df.round(6).to_string(index=False))

# 7. Display the 15 most frequent words under each estimate
def show_top_words(probabilities, method_name, top_n=15):
    top_indices = np.argsort(probabilities)[::-1][:top_n]

    table = pd.DataFrame({
        "Word": feature_names[top_indices],
        "Observed Count": category_word_counts[top_indices],
        "Estimated Probability": probabilities[top_indices]
    })

    print(f"\nTop {top_n} words using {method_name}:")
    print(table.round(6).to_string(index=False))
```

```
show_top_words(mle_probabilities, "MLE")

for alpha in prior_values:
    show_top_words(
        map_estimates[alpha],
        f"MAP (Dirichlet alpha={alpha})"
    )

# 8. Check that every estimated distribution sums to 1
print("\nPROBABILITY SUM CHECK")
print("MLE probability sum:", mle_probabilities.sum())

for alpha, probabilities in map_estimates.items():
    print(f"MAP alpha={alpha} probability sum:",
          probabilities.sum())

print("\nExperiment completed successfully.")
```

```
Loading 20 Newsgroups dataset...
Total documents: 11314
Number of categories: 20
Document-term matrix shape: (11314, 3000)
Vocabulary size: 3000

Selected category: sci.space
Number of documents: 593
Total word occurrences: 40862

MLE estimation completed.
MAP estimation completed for all prior values.

COMPARISON OF MLE AND MAP ESTIMATES
Method Prior alpha  Zero probabilities  Entropy (bits)  L1 distance from M
LE
   MLE        None                 508       10.273799              0.0000
00
   MAP         1.0                 508       10.273799              0.0000
00
   MAP         1.1                   0       10.297887              0.0071
25
   MAP         2.0                   0       10.457512              0.0668
64
   MAP        10.0                   0       11.058669              0.3889
54

Top 15 words using MLE:
     Word  Observed Count  Estimated Probability
    space             989               0.024203
     nasa             374               0.009153
   launch             267               0.006534
     like             222               0.005433
    earth             222               0.005433
     data             216               0.005286
    orbit             201               0.004919
     time             197               0.004821
  shuttle             192               0.004699
     just             189               0.004625
satellite             187               0.004576
    lunar             182               0.004454
     moon             168               0.004111
      new             158               0.003867
  program             156               0.003818

Top 15 words using MAP (Dirichlet alpha=1.0):
     Word  Observed Count  Estimated Probability
    space             989               0.024203
     nasa             374               0.009153
   launch             267               0.006534
     like             222               0.005433
    earth             222               0.005433
     data             216               0.005286
    orbit             201               0.004919
     time             197               0.004821
  shuttle             192               0.004699
     just             189               0.004625
satellite             187               0.004576
    lunar             182               0.004454
     moon             168               0.004111
```

```
      new             158               0.003867
  program             156               0.003818

Top 15 words using MAP (Dirichlet alpha=1.1):
     Word  Observed Count  Estimated Probability
    space             989               0.024029
     nasa             374               0.009088
   launch             267               0.006489
     like             222               0.005396
    earth             222               0.005396
     data             216               0.005250
    orbit             201               0.004886
     time             197               0.004788
  shuttle             192               0.004667
     just             189               0.004594
satellite             187               0.004545
    lunar             182               0.004424
     moon             168               0.004084
      new             158               0.003841
  program             156               0.003792

Top 15 words using MAP (Dirichlet alpha=2.0):
     Word  Observed Count  Estimated Probability
    space             989               0.022571
     nasa             374               0.008550
   launch             267               0.006110
     like             222               0.005084
    earth             222               0.005084
     data             216               0.004947
    orbit             201               0.004605
     time             197               0.004514
  shuttle             192               0.004400
     just             189               0.004332
satellite             187               0.004286
    lunar             182               0.004172
     moon             168               0.003853
      new             158               0.003625
  program             156               0.003579

Top 15 words using MAP (Dirichlet alpha=10.0):
     Word  Observed Count  Estimated Probability
    space             989               0.014706
     nasa             374               0.005644
   launch             267               0.004067
     like             222               0.003404
    earth             222               0.003404
     data             216               0.003316
    orbit             201               0.003095
     time             197               0.003036
  shuttle             192               0.002962
     just             189               0.002918
satellite             187               0.002888
    lunar             182               0.002815
     moon             168               0.002608
      new             158               0.002461
  program             156               0.002431

PROBABILITY SUM CHECK
MLE probability sum: 1.0
MAP alpha=1.0 probability sum: 1.0
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```
MAP alpha=1.1 probability sum: 1.0
MAP alpha=2.0 probability sum: 1.0
MAP alpha=10.0 probability sum: 0.9999999999999999

Experiment completed successfully.
```

In [ ]: